

MPS-P Series Programmable DC Power Supply SCPI Protocol Programming Manual

Case sensitivity

SCPI commands are case-insensitive: you can use uppercase, lowercase, or any combination of both.

For example: *RST = *rst

*IDN? = *idn?

*RCL = *rcl

Long and short

The SCPI command word can be sent in long form, short form, or a combination of both. This article provides the long form, while the short form is represented by uppercase characters.

for instance :

SYSTem: LOCal	Long
SYST: LOC	Short Form
SYSTem: LOC	combination of long and short
SYST: LOCal	combination of long and short

Note: Each command word must be in long or short form, not in a hybrid format.

For example: SYSTe: LOCa is invalid, this command will not be executed.

command format

In command syntax, most commands (and certain parameters) are written in mixed case. Capital letters denote abbreviations. For shorter lines, abbreviated forms are acceptable. For better readability, full-form commands are recommended. For example, both "VOLT" and "VOLTAGE" are valid formats. Capitalization is optional, so "VOLTAGE", "volt", and "Volt" are all acceptable. Other formats like "VOL" and "VOLTAG" are invalid and will not execute.

1. The curly braces ({}) contain parameter options for the given command string. The curly braces are not sent with the command string.
2. Vertical bar (|) separates multiple parameter options for a given command string. For example, in the command above, {0|1|OFF|ON} means you can specify "0", "1", "OFF", or "ON". The vertical bar is not included with the command string.
3. The angle brackets (<>) indicate that a value must be specified for the parameter within them. For example, VOLTage {<voltage value>}—angle brackets are not included in the command string. You must provide a value for the parameter. For example: VOLT 1.23
4. Colon (:) separates command keywords from the next level of keywords. For example: SYSTem: LOCal
5. Question mark (?) You can query the current value of a parameter by adding a question mark (?) to the command. For example: MEASure: VOLTage?
6. Space You must use a blank character, [TAB], or [space] to separate the parameter from the command keyword.
7. End character The command string sent to the instrument must end with a \r\n (0X0D,0X0A) character. The command string always resets the current SCPI command path to the root level.



remote interface connection

The power supply can be connected to the RS-232 interface via a level conversion circuit using the DB9 connector on the back panel. The following content will guide you on controlling the power output through a PC.

Communication Settings

Before performing communication operations, ensure the power supply matches the following PC parameters.

Baud rate: 9600

Data bits: 8

Stop position: 1

Verification: None

IEEE 488.2 general command

***IDN?**

This query command reads the power supply identifier string.

Return the following parameters: manufacturer name, product model, hardware version number, and software version number.

***RST**

This command resets the power to factory settings.

***SAV {<Address >}**

This command saves the current parameters to the specified address (0~9)

Example: *SAV 1

***RCL {<Address >}**

This command calls the parameter at the specified address (0~9).

Example: *RCL 1

SAV{<Address >}

This command saves the current parameters to the specified address (0~9)

Example: SAV2

RCL{<Address >}

This command calls the parameter at the specified address (0~9).

Example: RCL2

SYSTem order

The SYSTem command is used to set and query system-related status

SYSTem:LOCal

This command sets the power to local operation mode.

SYSTem:REMOte

This command sets the power to remote operation mode.

SYSTem: BEEPer {0 | 1 | OFF | ON}

This command enables or disables the power on/off sound.

Example: SYST: BEEP OFF "Disable the power-off notification sound"

SYST: BEEP 1 "Power-on alert"

SYSTem :BEEPer?

This command checks the power status notification sound.

Example: SYST: BEEP?

Return parameter: 0 (disable notification sound) | 1 (enable notification sound)

SYSTem: SENSE {0 | 1 | OFF | ON}

This command enables or disables remote power compensation.

Example: SYST: SENSE OFF "Disable remote compensation"

SYST:SENSE 1 "Enable remote compensation"

SYSTem: SENSE?

This command queries the remote power compensation status.

Example: SYST: SENSE:

Return parameter: 0 (disable remote compensation) | 1 (enable remote compensation)

SYSTem :ERRor?

This command queries power error information and clears error flags.

Example: SYST: ERR?

MEASure command

The MEASure command queries the actual output voltage and current values of the power supply.

MEASure: VOLTage?

This command queries the actual output voltage of the current power channel

Example: MEAS: VOLT?

Return parameter: actual voltage value (X.XXX)

MEASure: CURRent?

This command queries the actual output current value of the current power channel

Example: MEAS: CURR?

Return parameter: actual current value (X.XXX)

OUTPut order

The OUTPut command is used to configure and check the power output and its duration.

OUTPut {<0 | 1 | OFF | ON>}

This command enables or disables the output status of all power channels.

Example: OUTP OFF "Disable power output"

OUTP 1 "Enable power output"

OUTPut?

This command queries the output status of all power channels

example : OUTP?

Return parameter: 0 (disabled output) | 1 (enabled output)

VOLTage command

The VOLTAGE command is used to set and query voltage and overvoltage protection values.

VOLTage {<voltage>}

This command sets the output voltage value of the power supply.

Example: VOLT 12.345

Set the voltage to 12.345V

VOLTage?

This command queries the power supply set voltage value

example : VOLT?

Return parameter: voltage set value (X.XXX)

VOLTage:MINimun {<voltage>}

This command sets the minimum power output voltage.

Example: VOLT: MIN 3

Set the minimum voltage to 3A

VOLTage:MINimun?

This command queries the minimum power output voltage.

Example: VOLT: MIN?

Return parameter: Minimum power supply voltage (XX.XXX)

VOLTage: MAXimun {<voltage>}

This command sets the maximum power output voltage.

Example: VOLT: MAX 32

Set the voltage to the maximum value of 32V

VOLTage: MAXimun?

This command queries the maximum output voltage of the power supply.

Example: VOLT: MAX?

Return parameter: Maximum power supply voltage (XX.XXX)

VOLTage: PROTection {<ovp>}

This command sets the power overvoltage protection threshold.

Example: VOLT: PROT 12.3

Set the power supply overvoltage limit to 12.3V

VOLTage: PROTection?

This command queries the power supply's overvoltage protection threshold.

Example: VOLT: PROT?

Return parameter: Power overvoltage protection threshold (X.XXX)

VOLTage: PROTection:STATe {<0|1|ON|OFF>}

This command sets the power overvoltage protection status.

Example: VOLT: PROT: STAT OFF

Turn off the power overvoltage protection

VOLTage: PROTection:STATe?

This command checks the power overvoltage protection status

Example: VOLT: PROT: STAT?

Return parameter: 0 (disabled output) | 1 (enabled output)

CURRent command

The CURRent command is used to set and query current values and overcurrent protection status.

CURRent {<current>}

This command sets the power output current value.

Example: CURR 2.345

Set the power supply output current to 2.345A

CURRent?

This command queries the power supply set current value

example : CURR?

Return parameter: Power current set value (X.XXXX)

CURRent:MINimum

This command sets the minimum power output current.

Example: CURR: MIN 3

Set the current to a minimum of 3A

CURRent:MINimum?

This command queries the minimum power output current.

Example: CURR: MIN?

Return parameter: minimum power current (X.XXXX)

CURRent: MAXimum

This command sets the maximum power output current.

Example: CURR: MAX 5

Set the current to a maximum of 5A

CURRent: MAXimum?

This command queries the maximum power output current.

Example: CURR: MAX?

Return parameter: Maximum power current (X.XXXX)

CURRent: PROTection {<ocp>}

This command sets the power overcurrent protection threshold.

Example: CURR: PROT 3

Set the power supply overcurrent limit to 3A

CURRent: PROTection?

This command queries the power supply's overcurrent protection threshold.

Example: CURR: PROT?

Return parameter: Power overcurrent protection value (X.XXXX)

CURRent: PROTection:STATe {<0|1|ON|OFF>}

This command sets the power overcurrent protection status

Example: CURR: PROT: STAT ON

Enable power overcurrent protection

CURRent: PROTection:STATe?

This command checks the power overcurrent protection status

Example: CURR: PROT: STAT?

Return parameter: 0 (disabled output) | 1 (enabled output)